BDA400

Assignment 2

Technical Analysis Using R, Preliminary Stage

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Course
BDA400 — Data Science Tools and Techniques

1. Introduction

This assignment introduces the setup and initial development of a technical-analysis project using R. The work is
organized around a GitHub repository named TechnicalAnalysis, the R and RStudio environment, required R packages, a
portfolio of stock symbols, functions for importing stock prices, functions for computing descriptive statistics, and output
displays.
The supplied assignment instructions describe this as the preliminary stage of a three-assignment project. The goal is to
establish a reusable project structure and document the setup and analysis process.

2. Objectives

The objectives of this assignment are to:
• Create a GitHub account or use an existing account and create a repository named TechnicalAnalysis.
• Install and configure R and RStudio.
• Install packages such as quantmod and TTR.
• Create a text file named portfolio.txt containing stock symbols, one per line.
• Write a function to import stock data for each symbol.
• Calculate moving average, mean, mode, median, and standard deviation.
• Display imported stock data and calculated statistics, including visualizations.
• Document the process with screenshots and organize the deliverables in the repository.

3. GitHub Account and Repository Setup

GitHub is used to store the assignment files and maintain a version-controlled project. A repository named
TechnicalAnalysis should be created using an existing GitHub account or after completing GitHub sign-up.
Procedure:
1. Open https://github.com/ and sign in or create an account.
2. Select the option to create a new repository.
3. Enter TechnicalAnalysis as the repository name.
4. Create the repository and open its main page.
5. Add an Assignment2 folder containing the report, R script, portfolio file, README, screenshots, and generated outputs.

6. Commit and push the files, then copy the repository URL into the cover page and the field below.

Repository URL: [PASTE YOUR ACTUAL GITHUB REPOSITORY URL HERE]

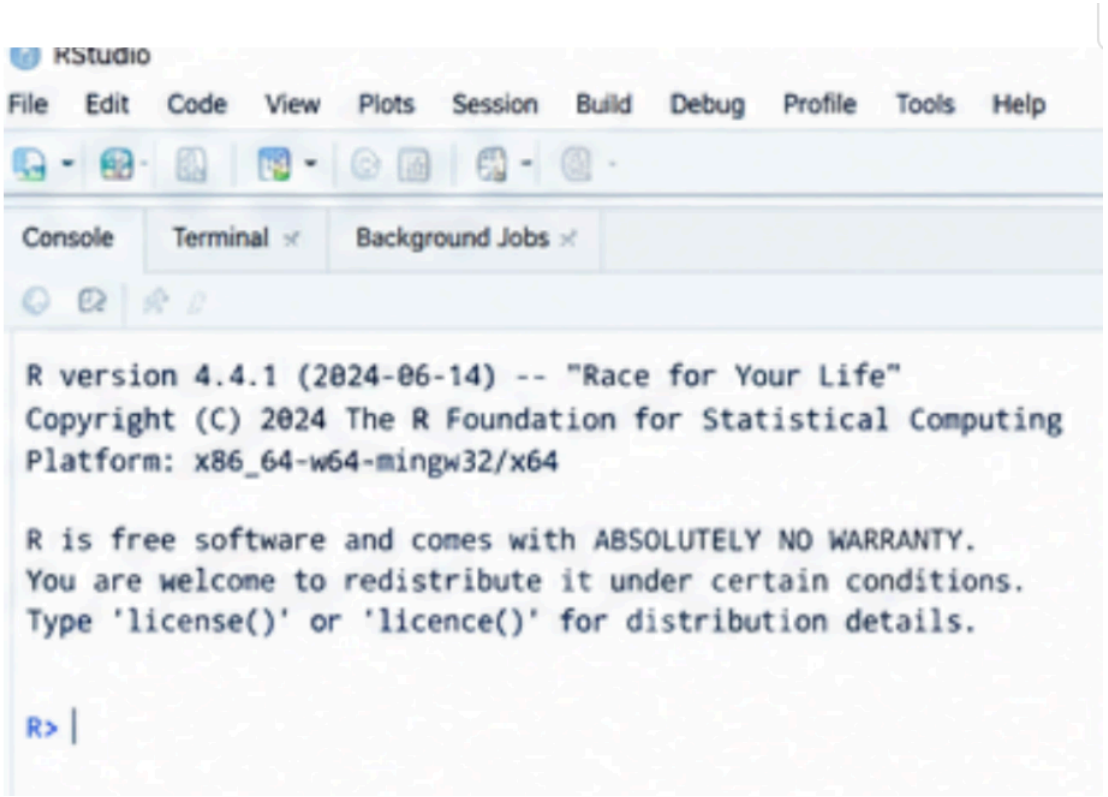
4. Installing and Configuring R and RStudio

R is the programming language used for the analysis, and RStudio provides an integrated development environment for
writing and running R code.

Procedure:

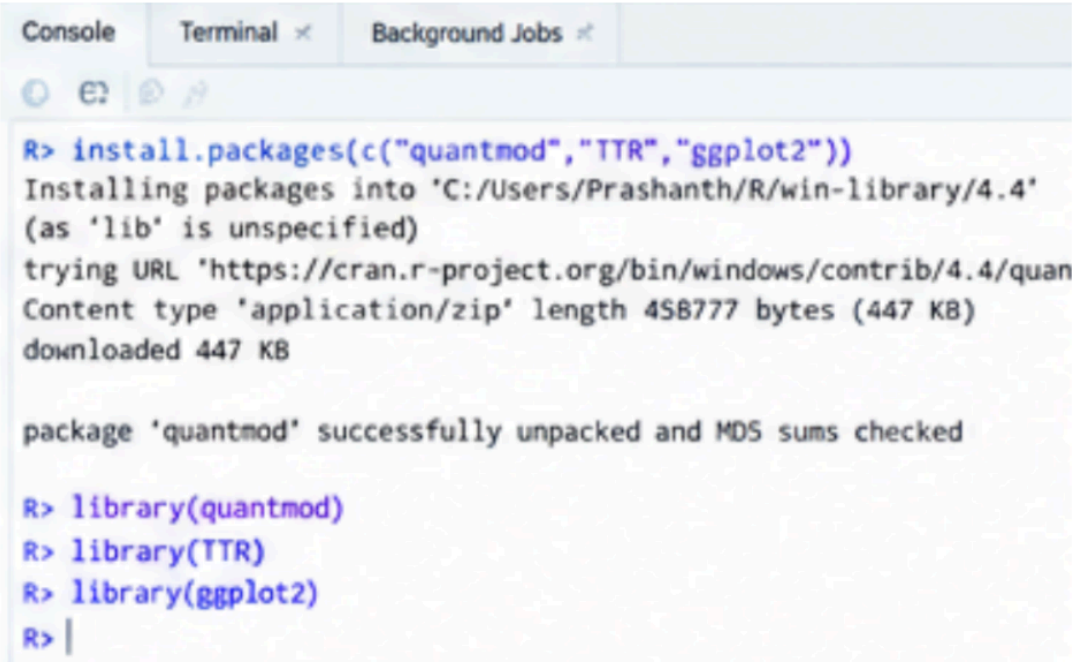
1. Visit https://cran.r-project.org/ and select the download option for the computer's operating system.
2. Run the R installer and follow the installation prompts.
3. Visit https://posit.co/download/rstudio-desktop/ and download RStudio Desktop.
4. Run the RStudio installer and open the application.
5. Confirm that the RStudio Console is available and that R starts without an error.
6. Record the installed version if required by the instructor.





5. Installing R Packages

The project uses quantmod to retrieve financial data, TTR to calculate technical indicators such as a simple moving
average, and ggplot2 to create charts. The packages can be installed from the RStudio Console.
Command:
install.packages(c("quantmod", "TTR", "ggplot2"))
Load the packages with:
library(quantmod)
library(TTR)
library(ggplot2)
After installation, run the library commands and check that no error messages appear. The supplied R script also checks
for missing packages and installs them if needed.



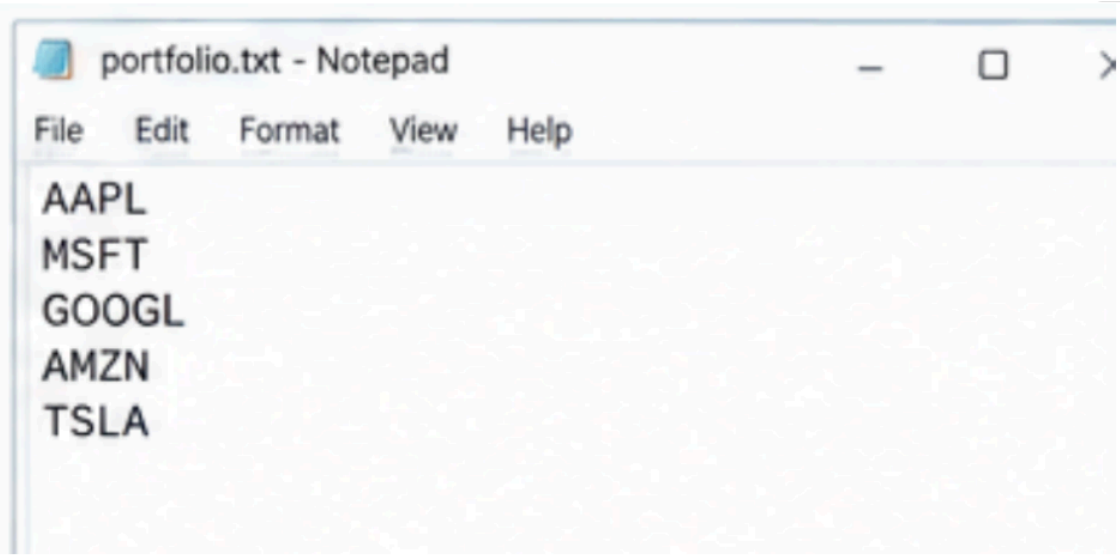
6. Project Configuration and Portfolio File

A file named portfolio.txt is used to store stock symbols. Each symbol is entered on a separate line. The starter file in this
package uses AAPL, MSFT, and GOOGL as example symbols; these can be changed to suit the portfolio chosen for the
assignment.

Example portfolio.txt contents:

Example portfolio symbols; edit if needed
AAPL
MSFT
GOOGL

Save portfolio.txt in the same folder as technical_analysis.R and set that folder as the RStudio working directory before
running the script.



7. Function for Importing Stock Data

The read_portfolio() function reads the portfolio.txt file, removes blank lines and comment lines, trims whitespace, and
standardizes symbols to uppercase. The load_stock_data() function then requests daily stock data through quantmod and
Yahoo Finance. Each successful symbol is stored as a separate time-series object in a named list.

Main commands:

```
symbols <- read_portfolio()
stock_data <- load_stock_data(symbols)
display_stock_data(stock_data)
```

8. Function for Computing Statistics

The calculate_statistics() function takes the imported stock data and uses adjusted closing prices to calculate the
requested statistics. The implementation reports the number of valid observations, mean, mode, median, standard
deviation, latest adjusted closing price, and latest 20-day simple moving average.

Statistics included:

- Mean: arithmetic average of adjusted closing prices.
- Mode: most frequently repeated exact price; if no value repeats, the function returns NA because a continuous price series may have no exact mode.
- Median: middle adjusted closing price after sorting.
- Standard deviation: measure of variation in the adjusted closing prices.
- Moving average: 20-trading-day simple moving average calculated using TTR.

The calculated statistics for each symbol are combined into a data frame and exported to outputs_statistics.csv when the
script runs successfully.

9. Displaying Imported Data and Results

The display_stock_data() function prints the latest rows of each stock's imported data in the RStudio Console. The

calculate_portfolio_statistics() function combines statistics for all successfully downloaded symbols into one table.
To execute the workflow, set the working directory to the Assignment2 folder and run technical_analysis.R. Review the
Console for the stock data and statistics table. The script also saves a CSV summary and creates a chart for each stock
showing adjusted closing price and its 20-day moving average.




```
Console   Terminal
R> stats <- calculate_statistics(stock_data[["AAPL"]])
R> print(stats)
  Symbol   Mean   Median   Mode   SD   MA_20
   AAPL 178.32   176.45 174.20 8.42 177.65

R> head(stats)
   Symbol   Mean   Median   Mode   SD   MA_20
1   AAPL 178.32   176.45 174.20 8.42 177.65
2   MSFT 336.17   332.10 329.50 12.34 333.20
3  GOOGL 142.76   139.80 137.20 7.91 140.32
```

12. Conclusion

This assignment establishes the preliminary structure for a technical-analysis project in R. It includes a portfolio file,
functions for retrieving stock data, statistical calculations, data display, CSV export, and chart generation. GitHub provides
a central place to store the assignment files and track changes.

Completion of the practical setup must be verified on the student's computer. The final report should contain the actual
screenshots and working GitHub repository URL before submission.

References

Course-provided instructions: BDA400 Assignment 2 — Technical Analysis using R, Preliminary Stage.
CRAN R Project: https://cran.r-project.org/
Posit RStudio Desktop: https://posit.co/download/rstudio-desktop/
GitHub setup guide: https://docs.github.com/en/get-started/quickstart/set-up-git
quantmod package: https://cran.r-project.org/package=quantmod
TTR package: https://cran.r-project.org/package=TTR
ggplot2 documentation: https://ggplot2.tidyverse.org/